

EXP12: Tic Tac Toe Game

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Aim:

To develop a Python program that implements the Tic Tac Toe game between two players, so that the board is updated after every move and a winner or draw is correctly identified.

Algorithm:

Step 1: Initialize an empty 3x3 board represented as a list of 9 cells.

Step 2: Display the current state of the board.

Step 3: Allow the current player (X or O) to make a move at a chosen position.

Step 4: Update the board with the player's symbol at that position.

Step 5: Check all rows, columns and diagonals to see if the current player has a winning combination.

Step 6: If a player wins, display the winner and stop the game.

Step 7: If no player has won and the board is full, declare the game a draw.

Step 8: Otherwise, switch to the other player and repeat Steps 2 to 7 until the game ends.

Source Code:

```
# Program to implement the Tic Tac Toe Game
```

```
def print_board(board):
    for i in range(0, 9, 3):
        print(" | ".join(board[i:i + 3]))
        if i < 6:
            print("-" * 9)
    print()

def check_winner(board, player):
    wins = [(0, 1, 2), (3, 4, 5), (6, 7, 8),
            (0, 3, 6), (1, 4, 7), (2, 5, 8),
            (0, 4, 8), (2, 4, 6)]
    return any(board[a] == board[b] == board[c] == player for a, b, c in wins)

board = ['1', '2', '3', '4', '5', '6', '7', '8', '9']
moves = [(0, 'X'), (4, 'O'), (1, 'X'), (5, 'O'), (2, 'X')]

print("Initial Board:")
print_board(board)

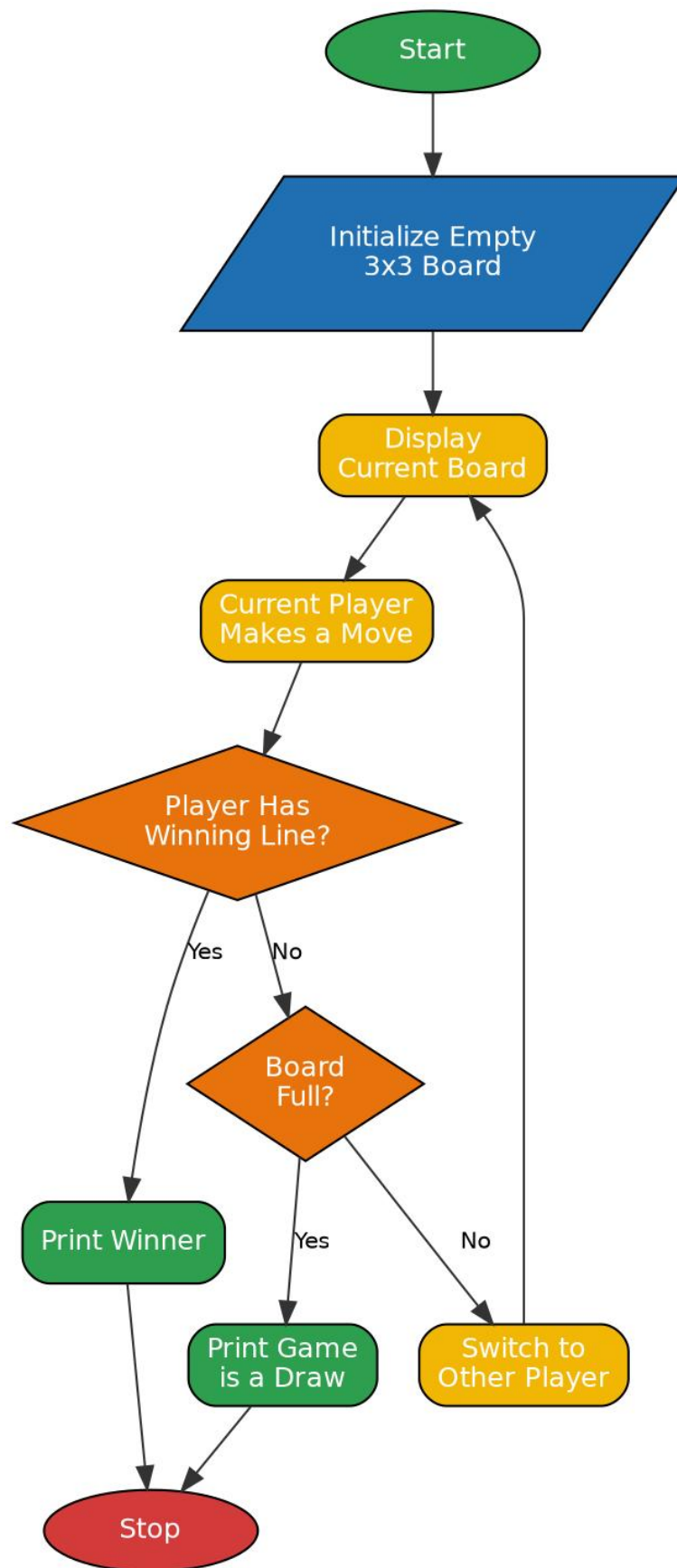
winner = None
for position, player in moves:
    board[position] = player
```

```
print(f'Player {player} plays at position {position + 1}')
print_board(board)

if check_winner(board, player):
    winner = player
    break

if winner:
    print(f'Player {winner} wins the game!')
else:
    print("The game ended in a draw.")
```

Flowchart:



Output:

```
>>> ===== RESTART: C:/Users/yanar/Documents/CSA1717/EXP12.py =====
Initial Board:
1 | 2 | 3
-----
4 | 5 | 6
-----
7 | 8 | 9

Player X plays at position 1
X | 2 | 3
-----
4 | 5 | 6
-----
7 | 8 | 9

Player O plays at position 5
X | 2 | 3
-----
4 | O | 6
-----
7 | 8 | 9

Player X plays at position 2
X | X | 3
-----
4 | O | 6
-----
7 | 8 | 9

Player O plays at position 6
X | X | 3
-----
4 | O | O
-----
7 | 8 | 9

Player X plays at position 3
X | X | X
-----
4 | O | O
-----
7 | 8 | 9

Player X wins the game!
>>>
```

Result:

Thus the program to implement the Tic Tac Toe game was written, executed successfully and the output was verified.